

matic. It reads a set of gschem schematic files, and creates a spreadsheet showing all components in rows, with the associated component attributes listed in the columns.

*libgeda*. It has libraries for gschem, gnetlist and gsymcheck.

*gsch2pcb*. It offers forward annotation from schematic to layout using PCB.

*gschem*. It is a schematic capture program.

Homepage: [www.gpleda.org](http://www.gpleda.org)

## GNU Octave

GNU Octave is a high-level language primarily intended for numerical computations. It provides a convenient command-line interface for solving linear and non-linear problems numerically, and for performing other numerical experiments using a language that is mostly compatible with MATLAB. It may also be used as a batch-oriented language.

The program runs under most Unix and Unix-like operating systems including Mac OS X and Microsoft Windows. It also provides extensive graphics capabilities for data visualisation and manipulation.

Microsoft version downloadable at: [http://sourceforge.net/projects/octave/files/Octave\\_Windows%20-%20MinGW/Octave%203.4.3%20for%20Windows%20MinGW%20Installer/](http://sourceforge.net/projects/octave/files/Octave_Windows%20-%20MinGW/Octave%203.4.3%20for%20Windows%20MinGW%20Installer/)

Homepage: [www.gnu.org/software/octave](http://www.gnu.org/software/octave)

## KiCad

KiCad is a multiplatform Open Source software distributed under the GNU GPL which is useful for everybody working in electronics design. It is a set of four software and a project manager—Eeschema (schematic entry), Pcbnew (board editor), GerbView Gerber viewer (photoplotter documents), CvpCb (footprint selector for components used in circuit design) and Kicad (project manager).

Kicad supports designing of electronics schematic diagrams and PCB artwork up to sixteen layers.

Microsoft version downloadable at:

[http://iut-tice.ujf-grenoble.fr/cao/KiCad-2011-11-27-RC1a-BZR3249-WinXP\\_full\\_with\\_components\\_doc\\_install.exe](http://iut-tice.ujf-grenoble.fr/cao/KiCad-2011-11-27-RC1a-BZR3249-WinXP_full_with_components_doc_install.exe)

Homepage: [http://kicad.sourceforge.net/wiki/Main\\_Page](http://kicad.sourceforge.net/wiki/Main_Page)

## KTechLab

KTechLab is an Open Source integrated development environment (IDE) for electronic and PIC microcontroller circuit design and simulation. It features an extensive circuit designer with auto-routing and simulation of all common electronic components and logic elements. KTechLab consists of several well-integrated components including a circuit simulator capable of simulating logic, linear devices and some non-linear devices.

Microsoft version downloadable at: [www.winkde.org/pub/kde/ports/win32/installer/kdewin-installer-gui-latest.exe](http://www.winkde.org/pub/kde/ports/win32/installer/kdewin-installer-gui-latest.exe)

Homepage: <http://sourceforge.net/projects/ktchlab/>

## Magic

Magic is a VLSI layout tool that is widely regarded as the easiest tool for circuit layout—even for people who ultimately rely on commercial tools for their product design flow.

Microsoft version downloadable at: <http://opencircuitdesign.com/cygwin/archive/magic-8.0.96v.tgz>

Homepage: <http://opencircuitdesign.com/magic>

## PCB

PCB is an Open Source software suite for PCB layout. It is an interactive PCB editor for the X Window system. It includes a rats nest feature, design rule check and can provide industry-standard RS-274-X (Gerber), NC drill and centroid data (X-Y data) output for use in the board fabrication and assembly process.

PCB offers high-end features such as an auto-router and trace optimiser. For custom requirements, it offers a plug-in application programming interface for inserting new functionality and using that functionality from

within the GUI as well as in scripts.

Microsoft version downloadable at: <http://sourceforge.net/projects/pcb/files/latest/download?source=files>

Homepage: <http://pcb.gpleda.org/index.html>

## Piklab

Piklab is an IDE for applications based on Microchip PIC and dsPIC microcontrollers similar to the MPLAB environment. It integrates with several compiler and assembler toolchains (like gputils, sdcc and c18) and with the GPsim simulator. It supports common programmers (serial, parallel, ICD2, Picket2 and PicStart+), ICD2 debugger and several bootloaders (Tiny, Picket2 and Demu).

Microsoft version downloadable at: <http://sourceforge.net/projects/piklab/files/newest%20release/0.15.10/piklab-prog-0.15.10-static.zip/download>

Homepage: <http://piklab.sourceforge.net>

## Quite Universal Circuit Simulator (Qucs)

Qucs is an integrated circuit simulator which allows the user to set up a circuit with GUI and simulate the large-signal, small-signal and noise behaviour of the circuit. After simulation, the simulation results can be viewed on a presentation page or window. The software aims to support all kinds of circuit simulation types, e.g., direct-current, alternating-current, S-parameter, harmonic balance analysis and noise analysis. Qucs supports a growing list of analogue and digital components as well as SPICE sub-circuits.

Microsoft version downloadable at: [http://sourceforge.net/project/showfiles.php?group\\_id=90337&package\\_id=192174](http://sourceforge.net/project/showfiles.php?group_id=90337&package_id=192174)

Homepage: <http://qucs.sourceforge.net>

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